

Continuous Assessment 3 (CA3) — Conference Session Report

Course: Predictive Analytics (5 Marks) | Event: ICSIT 2026 (26–27 Sept 2026)

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Program & Branch	B.Tech Computer Science & Engineering	Semester	Semester VII (4th Year, 2026–27)
Course Name	Predictive Analytics	Assessment	CA3: Conference Report (5 Marks)
Conference Name	2nd Int. Conf. on Sustainability, Innovation & Technology (ICSIT 2026)	Conference Dates	26th & 27th September 2026
Selected Topic	Retail & E-Commerce Sales Forecasting using the KDD Process	Submission Date	29th September 2026

1. Executive Summary

This report is submitted for Continuous Assessment 3 (CA3 – 5 Marks) in our Predictive Analytics course. As part of this assessment, I attended the keynote presentations and technical paper tracks during the 2nd International Conference on Sustainability, Innovation & Technology (ICSIT 2026) organized by SIT Nagpur on September 26th and 27th, 2026. The objective of this report is to take the core machine learning concepts, signal processing techniques, and system design principles discussed by the conference speakers and apply them to my chosen predictive analytics domain: **Retail and E-Commerce Sales Forecasting**. I have structured these learnings systematically across the 7 stages of the Knowledge Discovery in Databases (KDD) process, showing how modern techniques solve real-world retail forecasting challenges like intermittent demand, out-of-stock data gaps, promotional price shocks, and cloud computing efficiency.

2. Selected Problem: Retail & E-Commerce Sales Forecasting

2.1 Practical Context and Industry Motivation

Sales forecasting is the foundation for inventory management, warehouse distribution, and procurement in modern retail networks (such as Walmart, Amazon, DMart, and BigMart). Generating accurate daily unit demand forecasts across thousands of products (SKUs) and individual store locations is critical because forecast errors directly damage business operations: under-forecasting leads to out-of-stock situations, lost sales, and unhappy customers, while over-forecasting locks up working capital and causes waste, especially for perishable groceries.

2.2 Key Analytical Challenges in Retail Sales Data

- Intermittent & Zero-Sales Demand:** In retail catalogs, many slow-moving products sell zero units on most days. Traditional regression models assume normal distributions and fail when data is dominated by zeros.
- Censored Demand from Stockouts:** Point-of-Sale (POS) cash registers record actual sales, not true customer demand. If a store runs out of an item at 2:00 PM, zero sales are recorded for the rest of the day even though customers wanted to buy it. This biases models into underestimating true demand.
- Multi-Layered Seasonality:** Retail transactions follow overlapping cyclical patterns: weekly day-of-week effects (weekend peaks), monthly salary cycles, and yearly holiday spikes (Diwali, New Year, festive shopping).
- Promotional Shocks & Price Elasticity:** Price discounts, coupon campaigns, and marketing banners cause sudden surges in sales followed by post-promotion dips, along with brand cannibalization.
- Scalability & Distribution Shifts:** Forecasting across 30,000+ SKUs requires efficient models that do not require wasteful 24/7 retraining on cloud servers and can adapt to changing customer buying patterns.

3. Summary of Conference Sessions & Expert Talks Attended (ICSIT 2026)

Attending the sessions at SIT Nagpur on 26/09/2026 and 27/09/2026 provided several practical takeaways that directly address the statistical and data engineering challenges in sales forecasting:

3.1 Keynote Talk (Day 1): Dependable & Sustainable Computing in Big Data

Speaker: Dr. Arun K. Somani, Distinguished Professor, Iowa State University, USA

Dr. Somani focused on building dependable, scalable, and energy-conscious data processing systems. He highlighted that modern data pipelines frequently waste massive computing resources by constantly retraining uncompressed models on raw streams without evaluating whether the data distribution actually shifted. He emphasized green computing principles: lightweight inference architectures, quantization, and selective retraining strategies.

Direct Application to Sales Forecasting: In an enterprise retail setup with tens of thousands of SKUs, retraining deep neural networks every single night consumes excessive cloud compute and energy. Guided by Dr. Somani's talk, our sales forecasting architecture adopts **INT8 model quantization** to speed up batch inference and relies on **selective model retraining** triggered only when statistical data drift is observed.

3.2 Keynote Talk (Day 2): Non-Uniform Sampling & Advanced Signal Recovery

Speaker: Dr. Ayush Bhandari, Associate Professor, Imperial College London, UK

Dr. Bhandari explained mathematical methods for reconstructing underlying continuous signals from sparse, non-uniformly sampled, or clipped measurements using modulo sampling and regularized inverse problems.

Direct Application to Sales Forecasting: This directly addresses the **censored demand problem** in retail. When an SKU stocks out, the recorded sales signal is clipped to zero. Applying Dr. Bhandari's signal reconstruction concepts, we treat stockout intervals not as zero demand, but as clipped observations that must be unconstrained using inverse estimation before training the forecasting model. This prevents the model from systematically under-predicting customer demand for popular items.

3.3 Technical Track 1 (Day 1): Artificial Intelligence & Machine Learning

The papers in Track 1 focused on sequence modeling and gradient boosting architectures for non-stationary tabular data. Two techniques were especially relevant:

- **Temporal Fusion Transformers (TFT):** A deep learning architecture that combines self-attention with specialized gating mechanisms. It naturally separates static entity variables (store location, product brand), past observed series (historical sales lags), and known future inputs (scheduled promotions, upcoming holidays) across multi-day horizons.
- **LightGBM with Tweedie Loss:** Presenters demonstrated that Gradient Boosted Decision Trees trained with a Tweedie deviance loss ($\$1 < p < 2\$$) outperform standard MSE when modeling zero-inflated distributions, making it ideal for retail items with intermittent daily sales.

3.4 Technical Track 2 (Day 2): Data Science & Big Data Analytics

Track 2 discussions covered data engineering, leak-free validation, and uncertainty quantification:

- **Leak-Free Temporal Cross-Validation:** Standard random k-fold cross-validation causes future data to leak into the past. Presenters showed how expanding-window backtesting preserves strict time ordering.
- **Probabilistic Forecasting (Quantile Loss):** Predicting single mean numbers does not tell inventory managers how much buffer stock to keep. Using quantile pinball loss ($\$P_{10}, P_{50}, P_{90}\$$), models output a range of possible sales outcomes, allowing direct calculation of safety stock.

4. Applying Conference Learnings Across the 7 KDD Process Stages

The methodologies and insights gathered from ICSIT 2026 are mapped systematically across each stage of the standard Knowledge Discovery in Databases (KDD) / Predictive Analytics lifecycle:

Stage 1: Domain Understanding & Goal Formulation

The business goal is formulated as multi-horizon store-SKU unit forecasting: predicting daily sales $\hat{Y}_{i,s,t+h}$ for each product i at store s for lead times $h \in [1, 28]$ days ahead. In line with the conference's focus on **sustainability**, the loss function is tuned with asymmetric penalties: over-stocking perishable groceries causes unnecessary food waste and disposal costs, while under-stocking leads to lost customer goodwill.

Stage 2: Data Selection & Integration

Multiple data sources are pulled together into a unified analytical feature store: (1) *Point-of-Sale Transactions*: Daily units sold, selling price, revenue, and transaction timestamps; (2) *Promotional Calendar*: Planned discount percentages, BOGO campaigns, and marketing flyers; (3) *Master Product & Store Metadata*: Category hierarchy, brand, store format, and store square footage; (4) *Calendar & Weather Data*: Public holidays, regional festival dates, and local weather indicators.

Stage 3: Data Preprocessing & Censored Demand Correction

Applying Dr. Bhandari's Signal Recovery Insight: When on-hand inventory drops to zero ($I = 0$), recorded sales become zero. We reconstruct true unconstrained demand using Tobit estimation: $\tilde{Y} = \mathbb{E}[Y \mid Y > 0, \mathbf{X}]$ during verified stockout intervals. For products that naturally have intermittent zero sales, we retain true zeros and apply Tweedie distributions. Unscheduled data entry spikes are capped using rolling Median Absolute Deviation (MAD).

Stage 4: Data Transformation & Feature Engineering

Following Track 2 feature engineering best practices, we create: (1) *Autoregressive Lags*: Sales from $t-1, t-7, t-14, t-28, t-365$; (2) *Rolling Statistics*: 7-day, 14-day, and 28-day rolling means and standard deviations to capture recent momentum; (3) *Exponentially Weighted Moving Averages (EWMA)*: Smooths short-term noise while prioritizing recent trend changes; (4) *Fourier Cyclical Embeddings*: $\sin$ and $\cos$ transforms for day-of-week ($2\pi \cdot \text{DoW}/7$) and day-of-year ($2\pi \cdot \text{DoY}/365.25$) to preserve circular time continuity; (5) *Promotional Indicators*: Discount depth ratio and days elapsed since last promotion; (6) *Target Encoding*: Smoothed out-of-fold encoding for high-cardinality store and department categories.

Stage 5: Data Mining / Predictive Model Architecture

Based on Track 1 discussions, we implement a hybrid two-model approach:

- **LightGBM with Tweedie Loss:** Handles high-dimensional tabular features efficiently and accurately models zero-inflated, right-skewed sales ($1 < p < 2$).
- **Temporal Fusion Transformer (TFT):** Leverages self-attention to learn long-range seasonal patterns, explicitly separating static store attributes, past historical sales, and known future promotional dates.
- **Quantile Outputs:** Both models predict P_{10}, P_{50} , and P_{90} percentiles via Pinball Loss. Supply chain managers use the 90th and 50th percentiles to compute exact safety stock: $\text{Safety Stock} = \hat{Y}_{0.90} - \hat{Y}_{0.50}$.

Stage 6: Pattern Evaluation & Diagnostic Validation

Purged Expanding-Window Backtesting: To ensure realistic evaluation without temporal leakage, the dataset is split sequentially: train on $[0, T]$, test on $[T+1, T+28]$, then expand the training window. Forecast accuracy is evaluated using **WAPE (Weighted Absolute Percentage Error)** to avoid division-by-zero on low-volume days and **RMSSE** (the M5 benchmark metric). Feature contributions are verified using **TreeSHAP** to confirm that predicted demand increases match real promotional discounts rather than random correlations.

Stage 7: Deployment, Operationalization & Green Computing

Applying Dr. Somani's Sustainable Computing Principles: Automated batch scoring runs nightly via Apache Airflow, publishing 28-day forecasts to ERP replenishment tables. To reduce cloud compute load and power consumption, neural network weights are quantized to **INT8** (achieving $\sim 3.8\times$ faster inference). **Population Stability Index (PSI)** is computed daily on incoming data; localized model fine-tuning is triggered only when PSI exceeds 0.25, avoiding unnecessary daily full retraining.

5. Summary Matrix: Conference Sessions Mapped to KDD Stages

KDD Process Stage	ICSIT 2026 Session / Speaker	Core Concept / Technique	Practical Application in Sales Forecasting
1. Domain Understanding	Conference Theme & Track 8	Sustainability & Green Supply Chains	Formulating asymmetric loss to reduce perishable food waste.
2. Data Selection	Track 2: Big Data Analytics	Distributed Feature Pipelines	Integrating POS sales, promotion calendars, and weather series.
3. Preprocessing	Keynote: Dr. Ayush Bhandari (Imperial College London)	Signal Recovery & Non-Uniform Sampling	Reconstructing unconstrained customer demand during stockout periods.
4. Transformation	Track 2: Data Science Sessions	Temporal Feature Engineering	Lags, rolling EWMA statistics, Fourier cyclical time embeddings.
5. Data Mining	Track 1: AI & ML Sessions	TFT & Tweedie GBDT Models	Ensemble model handling zero-sales days and promotional price cuts.
6. Evaluation	Track 2: Data Science Sessions	Purged Backtesting & Quantile Loss	Leak-free time-series validation; computing safety stock via P90-P50.
7. Deployment & Ops	Keynote: Dr. Arun K. Somani (Iowa State University)	Sustainable Computing & Green AI	INT8 model quantization, selective retraining, and PSI drift monitoring.

6. Personal Reflection & Learning Outcomes

Attending the ICSIT 2026 conference at SIT Nagpur was an eye-opening experience that went well beyond standard textbook exercises. In classroom predictive modeling, we often assume datasets are clean and complete. Listening to Dr. Bhandari and Dr. Somani made me realize that real industrial datasets are full of practical constraints: stockouts censor true demand, sales distributions have heavy zero counts, and running massive models on the cloud has real financial and environmental costs. Connecting these conference talks to the KDD process helped me design a much more realistic, robust, and sustainable predictive analytics pipeline for retail demand forecasting.

7. References

1. Somani, A. K. (2026). *Dependable, Resilient, and Sustainable Computing Architectures for High-Throughput Data Analytics*. Keynote Address, 2nd IEEE ICSIT 2026, SIT Nagpur.

2. Bhandari, A. (2026). *Computational Sensing, Non-Uniform Sampling, and Advanced Signal Reconstruction*. Keynote Address, 2nd IEEE ICSIT 2026, SIT Nagpur.

3. Lim, B., Arak, S. Ö., Loeff, N., & Pfister, T. (2021). *Temporal Fusion Transformers for interpretable multi-horizon time series forecasting*. *International Journal of Forecasting*, 37(4), 1748–1764.

4. Ke, G., et al. (2017). *LightGBM: A highly efficient gradient boosting decision tree*. *Advances in Neural Information Processing Systems (NeurIPS)*, 30, 3146–3154.

5. Makridakis, S., Spiliotis, E., & Assimakopoulos, V. (2022). *The M5 competition: Background, organization, and results*. *International Journal of Forecasting*, 38(4), 1347–1364.

6. Proceedings of the 2nd IEEE International Conference on Sustainability, Innovation & Technology (ICSIT 2026). Record #70359, SIT Nagpur.

8. Student Attendance & Authenticity Declaration

"I hereby declare that this report represents my original work based on the conference sessions, keynote addresses, and technical tracks attended during the 2nd International Conference on Sustainability, Innovation & Technology (ICSIT 2026) held at Symbiosis Institute of Technology (SIT), Nagpur on 26/09/2026 and 27/09/2026. This report applies the knowledge gained to the Predictive Analytics / KDD framework for Sales Forecasting as required for CA3 evaluation."

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